

PeeCee



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
12 CLASSES (1 HOUR PER
CLASS)**

**MODE:
OFFLINE**

PeeCee is a microcontroller-based, hands-on learning kit designed for young learners. It uses a built-in controller with internal sensors and outputs such as microphone, light, temperature, touch inputs, LEDs, buzzer, motors and servo control.

Students learn logic, programming and automation by connecting inputs to outputs through simple programs instead of building snap-fit circuits.

Through guided activities and projects, learners understand how a microcontroller works as the brain of real electronic systems, while developing problem-solving, computational thinking and STEM fundamentals in a safe, child-friendly way.

COURSE CURRICULUM

PeeCee – Level 1

1) PeeCee: Seeing, Hearing and Believing

- Understanding how PeeCee senses light, sound, and touch.
- Introduction to inputs, outputs and real-time responses.

2) PeeCee Add-Ons: Next Cross and Elegant Light

- Exploring add-on components and their functions.
- Learning visual feedback and lighting control.

3) Scribbling Bot Construction and Basics of Switch & Motors

- Understanding motor movement and direction.
- Learning how switches control actions.

4) Scribbling Robot Programming & Algorithm

- Writing step-by-step programs.
- Understanding basic algorithms and sequences.

5) Rookie Programmer + Stand in Line + Disco Light Hanger Project

- Learning simple logic and patterns.
- Coordinating lights and actions using programs.

6) Walking Dog Construction

- Understanding mechanical movement.
- Connecting motors to motion concepts.

7) Programming of Walking Dog

- Controlling movement using logic and timing.

- Learning to start, stop and repeat actions.

8) Touch to Start + Automated Fan Project

- Using touch input to trigger actions.
- Understanding automation concepts.

9) Dependent on Light (LDR) + Laser Concept + Laser Burglar Alarm Project + Smart Night Lamp Project

- Understanding light sensing and response.
- Learning safety and automatic lighting systems.

10) Servo Motors + Variable Speed Fan Project + Unique Beginner Motor Project

- Understanding servo control and speed variation.
- Learning precise motor movement.

11) Temperature Sensor + Throne of PeeCee Project

- Understanding temperature sensing.
- Building temperature-based response systems.

12) Challenge Day

- Applying learned concepts to solve problems.
- Encouraging creativity and logical thinking.

PeeCee – Level 2

1) Moving Robo Car Construction

- Understanding structure and motion.
- Building a moving robotic system.

2) Programming of Moving Robo Car

- Controlling movement through logic.
- Learning direction and timing.

3) Ultrasonic Sensor Integration in Robo Car & Programming

- Understanding distance measurement.
- Obstacle detection and avoidance logic.

4) Gesture Sensor and Robotic Claw Construction

- Understanding gesture-based control.
- Building interactive robotic mechanisms.

5) Robotic Claw Programming

- Controlling grip and release actions.
- Learning precision and coordination.

6) Sound Activated Car: Mic, Buzzer & Programming

- Understanding sound input and output.
- Creating sound-based control systems.

7) Potentiometer & Servo Motor: Camera Pan System

- Understanding variable input control.
- Smooth angle and position control.

8) Proximity Sensor & Traffic Light Construction

- Understanding proximity sensing.
- Building signal-based control systems.

9) Traffic Light Programming

- Learning sequencing and timing.

- Understanding real-life signal logic.

10) Home Automation System Construction

- Understanding automated systems.
- Connecting sensors to appliances logically.

11) Home Automation Programming

- Writing programs for automation control.
- Understanding smart home concepts.

12) Challenge Day

- Solving real-world problems using PeeCee.
- Testing logic, accuracy and teamwork.

Class Structure

1. Introduction to PeeCee, Sensors and Outputs

- Learn about the microcontroller, built-in sensors, motors, lights and their functions.

2. Controller-Based Programming and Setup

- Program the PeeCee to control actions using a child-friendly interface under guidance.

3. Concept Understanding Through Actions

- Simple explanations of logic, sensors, and automation linked to visible outcomes.

4. Hands-on Activities and Projects

- Interactive programming tasks, robotics projects and challenge-based learning.

5. Safety, Discipline, and Real-Life Applications

- Safe handling of the kit and understanding how microcontroller systems are used in everyday technology.